

Exercise 4 - Theoretical Questions

Question 1: Lazy Lists

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Question 1.1: Lazy Lists

(a) Equivalence Criterion

Two lazy lists are considered equivalent if, after any finite number of applications of `tail`, either both lists are empty, or both lists have a defined `head` and these two `head` values are equal.

(b) Equivalence of `even-square-1` and `even-square-2`

Let

$$L_1 = \text{even-square-1}$$

and

$$L_2 = \text{even-square-2}.$$

We consider an arbitrary natural number T produced by `integers-from 0`.

In L_1 , the number T is first mapped to T^2 , and then the filter checks whether T^2 is even.

In L_2 , the number T is first filtered according to whether T is even, and only then it is mapped to T^2 .

We split into two cases.

If T is even, then T^2 is also even. Therefore, in L_1 , the value T^2 passes the filter. In L_2 , the value T passes the filter and is then mapped to T^2 . Thus, in both lists, the same value T^2 appears.

If T is odd, then T^2 is also odd. Therefore, in L_1 , the value T^2 does not pass the filter. In L_2 , the value T does not pass the filter before the mapping. Thus, in both lists, T contributes no value to the final lazy list.

Hence, for every natural number T , either both lists include the same value T^2 , or both lists exclude it. Since both `map` and `filter` preserve the original order of the remaining elements, both lazy lists produce the same sequence:

$$0, 4, 16, 36, 64, \dots$$

Therefore, according to the equivalence criterion defined in part (a), L_1 and L_2 are equivalent lazy lists.

Question 2: CPS

(a)

A procedure **func** of type $[A_1 * \dots * A_n \rightarrow T \mid \text{'fail}]$ and its Success-Fail-Continuations version **func\$** of type $[A_1 * \dots * A_n * [T \rightarrow T1] * [\text{Empty} \rightarrow T2] \rightarrow T1 \mid T2]$ are equivalent if and only if for every valid sequence $x_1, \dots, x_n$ and any continuation functions **success** and **fail**

1. $(\text{func } x_1 \dots x_n)$ will terminate and return value $\text{res} \neq \text{'fail} \iff (\text{func\$ } x_1 \dots x_n \text{ success fail})$ results in the evaluation of (success res) .
2. $(\text{func } x_1 \dots x_n)$ will terminate and return $\text{'fail} \iff (\text{func\$ } x_1 \dots x_n \text{ success fail})$ results in the evaluation of (fail) .
3. $(\text{func } x_1 \dots x_n)$ does not terminate $\iff (\text{func\$ } x_1 \dots x_n \text{ success fail})$ does not terminate.

2d

We will prove **get-value** and **get-value\$** equivalence using induction on $n = \text{len}(\text{assoc-list})$.

Base Case

On $n = 0$, $(\text{empty? assoc-list})$ evaluates to **true**. Therefore, **get-value** will terminate and return 'fail , and **get-value\$** will evaluate (fail) .

Proposition

For every $n > 0$, we will assume that for every **assoc-list** with $\text{len}(\text{assoc-list}) = n$, **get-value** and **get-value\$** are equivalent.

Induction Step

We will prove that $(\text{get-value assoc-list key})$ and $(\text{get-value\$ assoc-list key success fail})$ are equivalent for an **assoc-list** with $\text{len}(\text{assoc-list}) = n + 1$.

For easier reading, we will refer to $(\text{get-value assoc-list key})$ as **get-value** and $(\text{get-value\$ assoc-list key success fail})$ as **get-value\$**.

Since $\text{len}(\text{assoc-list}) > 0$, **get-value** will evaluate to:

```
(if (eq? key (car (car assoc-list)))
    (cdr (car assoc-list))
    (get-value (cdr assoc-list) key))
```

and `get-value$` will evaluate to:

```
(if (eq? key (car (car assoc-list)))  
    (success (cdr (car assoc-list)))  
    (get-value$ (cdr assoc-list) key success fail))
```

Case 1: `(eq? key (car (car assoc-list)))` evaluates to `#t`

In this case, `get-value` will evaluate to `(cdr (car assoc-list))`, and `get-value$` will evaluate to `(success (cdr (car assoc-list)))`. Therefore, they are equivalent.

Case 2: `(eq? key (car (car assoc-list)))` evaluates to `#f`

Here, `get-value` will return the evaluation of `(get-value (cdr assoc-list) key)`, and `get-value$` will evaluate `(get-value$ (cdr assoc-list) key success fail)`.

By assuming the induction hypothesis for `(cdr assoc-list)`, which has length n , it follows that `(get-value (cdr assoc-list) key)` and `(get-value$ (cdr assoc-list) key success fail)` are equivalent. Therefore, `get-value` and `get-value$` are equivalent.